MCNet: An Efficient CNN Architecture for Robust Automatic Modulation Classification

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***Abstract*— This letter proposes a cost-efficient convolutional neural network (CNN) for robust automatic modulation classification (AMC) deployed for cognitive radio services of modern communication systems. The network architecture is designed with several specific convolutional blocks to concurrently learn the spatiotemporal signal correlations via different asymmetric convolution kernels. Additionally, these blocks are associated with skip connections to preserve more initially residual information at multi-scale feature maps and prevent the vanishing gradient problem. In the experiments, MCNet reaches the overall 24-modulation classification rate of 93.59% at 20 dB SNR on the well-known DeepSig dataset.**

***Index Terms*— Automatic modulation classification, deep learning, convolutional neural network, skip connection.**

I. Introduction

NOWDAYS, respect to the emergence of multiple standards and innovative technologies for wireless communication, wherein radio signal should be encoded by a predefined modulation scheme according to the specification of transmission channel, spectrum analysis through the models of signal classification and modulation recognition plays a vital role in massive wireless communication systems like fifth generation (5G) [1]. Fundamentally, modulation recognition is the task of classifying the modulation type of a received radio signal, which is treated as a multi-class decision problem. With the increasing number of advanced modulation algorithms, the high recognition accuracy of modulation type is of great importance in noise and multipath fading conditions. By means of artificial intelligence (AI) algorithms [2], automatic modulation classification (AMC) is taken into account as a potential solution of efficient spectrum management.

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Most of the previous AMC works are typically categorized into the likelihood-based and feature-based groups. While the likelihood-based methods waste an expensive computational capacity for unknown parameter estimation of classification models [3], the feature-based approaches are mostly influenced by the experience and knowledge of signal feature engineering. The feature-based methods are commonly deployed for many wireless communication systems in practice thanks to the high reliability in combatting the harmful influence of multipath fading, but they pose the drawbacks of poorly represented feature and weakly discriminative machine learning (ML)-based classifier [4] if dealing with a high variety of challenging modulation formats such as 128APSK and 256QAM.

Deep learning (DL) architectures including recurrent neural networks (RNNs) and convolutional neural networks (CNNs) [5] are being exploited with outstanding achievements in wide-ranging fields from natural language processing and computer vision [6] to economics and bioinformatics [7]. Recently, DL is potentially studied for the tasks of signal classification and modulation recognition [8], wherein the facilitation of automatically powerful feature learning [9] enables a greater accuracy. In [10], an RNN-based AMC approach is studied to explicitly characterize the complex relations of time-correlated signal over Rayleigh fading channels along various additive noise conditions. Long short-term memory (LSTM) network [11], an advanced architecture of RNN, is specifically designed for learning the long-term dependencies in the time domain of varying-length modulation signals. Compared with RNNs and LSTM, CNNs capably extract more meaningful features of discrimination at multi-scale feature representations for the multi-class classification task [12]. A compact CNN involving several residual convolutional stacks [13] deployed for acquiring more relevant information at multi-level representational feature maps significantly improves the classification rate on the newest 24-modulation DeepSig dataset. Compared with ML, DL is superior in recognition performance without feature engineering knowledge [14]. However, by incorporating several ordinary structures of convolutional layers into a heavily deep network architecture, most of the current AMC methods are underperformed, which is caused by the ineffective representation of extracted feature maps and the expensive cost of trainable parameters. Consequently, they nearly fail to meet the baseline requirements of 5G services in terms of ultra-reliability and low-latency.

In this work, the aforementioned AMC problems are soundly handled by leveraging deep convolutional network (ConvNet). A cost-efficient and well-performed ConvNet is

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studied for robust modulation classification against channel impairments, which promisingly enhances the quality of 5G services. Regarding the network architecture, the main contributions are summarized as follows:

- Convolutional blocks designed with asymmetric kernels to reduce parameters and acquire more powerful features.
- Skip connections deployed for block-wise association to prevent the vanishing gradient problem and accordingly improve the performance of classification model.

II. Methodology

A. Fundamental of CNNs

Basically, a deep CNN architecture widely deployed for different fundamental computer vision tasks (for example, image classification, object detection, and semantic segmentation) should comprise numerous convolutional layers (conv) which are arranged in series and followed by one or more fully connected layers (FC) [15]. In general, the convolution of a two-dimensional (2-D) kernel and an input map, at any specific spatial coordinate (x, y), is the sum of dot products as follows

$$\text{conv}_{x,y} = \sum_i w_i v_i, \tag{1}$$

where w_i stands for the convolutional kernel weights and v_i refers to as the values of correspondingly spatial extent in the input map (or the output of the previous layer). The output of a convolutional layer is obtained by adding a scalar bias b

$$z_{x,y} = \text{conv}_{x,y} + b. \tag{2}$$

In ConvNets, a complete feature map (so-called activation map) is gained by passing the scalar $z_{x,y}$ through a nonlinear activation function g as follows

$$a_{x,y} = g\left(z_{x,y}\right). \tag{3}$$

As one of the most widely used functions in practice, rectified linear unit (ReLU) executes a threshold operation to each element z, where any value less than zero is scaled to zero

$$g\left(z\right) = \begin{cases} z & \text{if } z \geq 0, \\ 0 & \text{if } z < 0. \end{cases} \tag{4}$$

The ReLU layer does not change the size of its input. Throughout the architecture of ConvNet, spatial down-sampling allows to reduce the size of feature maps and eliminate redundant information via max- or average-pooling layers. Accordingly, a deeper network with more convolutional layers is assembled without the overgrowth of computational complexity.

B. MCNet - A Low-Cost CNN for Robust AMC

In many innovative communication systems, latency is one of the most important factors impacting on the service quality, which demands a well-performed AMC model in terms of classification rate and processing speed, especially if AMC is developed for services operating on edge computing platform. Hence, this section presents the cost-efficient architecture of a modulation classification convolutional neural network, called MCNet, for comprehensively taking into account the current challenges of modulation classification. Generally, the deep network is specifically designed with several convolutional blocks associated with skip connections as shown in Fig. 1, in which each block comprises various asymmetric convolutional layers. The proposed network is capable of analyzing the multi-scale spatiotemporal signal correlations exhaustively to promisingly improve the accuracy of modulation classification under poor conditions with the cheapest computational cost.

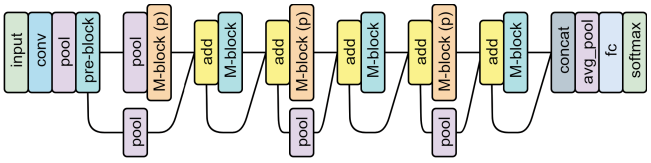
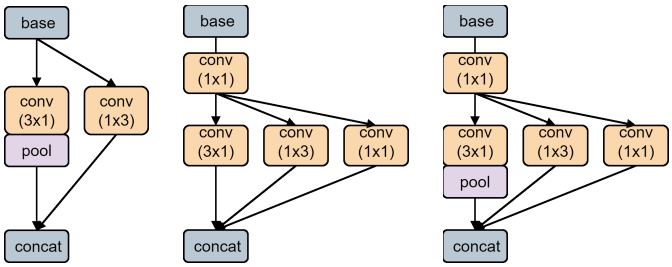


Fig. 1. Overall network architecture of MCNet involving multiple convolutional blocks connected via skip connections.



(a) Pre-block (b) M-block (c) M-block (p)

Fig. 2. Description of convolutional blocks developed in the MCNet. (a) the pre-block with dual convolutional flows of asymmetric filters 3×1 and 1×3; (b) the convolutional M-block with triple flows of convolutional filters 3×1, 1×3, and 1×1; and (c) the dimension-reduced version of M-block with an additional max-pooling layer.

The input layer is configured to facilitate with short-time signal partitions, so-called signal frames. For instance, each frame in the DeepSig dataset [13] is organized as the two-dimensional array with the size of 2×1024 (i.e., the I/Q modulation signal is sampled with $\ell = 1024$ elements). With 3×7 kernels, the first convolutional layer is able to cover generic features and further reduce spatial dimension with the stride of $(1, 2)$. After that, two layers of asymmetric convolutional kernels (for details, 3×1 and 1×3 corresponding to vertical and horizontal kernels) organized in parallel of a pre-block as illustrated in Fig. 2a are deployed to minimize the number of trainable parameters without the quality degradation of extracted features if compared with the use of 3×3 kernels [13]. It is worth noting that both the convolution and pooling operations are carried out with stride of $(1, 2)$ at the beginning to quickly diminish the horizontal dimension of feature maps and subsequently reduce the computational cost of several following layers.

As the core of MCNet, six convolutional blocks for learning more explicitly discriminative features, denoted M-block as in Fig. 2b, are next executed after down-sampling feature maps with a max-pooling layer. Concretely, each M-block comprises three convolutional layers of different kernels, particularly 3×1, 1×3, and 1×1 kernels arranged in parallel, at which all feature maps are concatenated in the depth dimension at the output of each block. The spatial dimension of feature

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maps is preserved at the output of M-block for the reason that all kernels are deployed with stride $(1, 1)$. Besides, another dimension-reduced version of M-block is given with a same formation, except a max-pooling operation is carried out following a 3×1 convolutional layer as shown in Fig. 2c. As a result, M-block with pooling halves in the horizontal dimension of the concatenated feature map. To be compatible with output of the pooling layer when performing depth concatenation, two remaining layers are conducted with stride of $(1, 2)$. Each block has two 1×1 convolutional layers: one for feature extraction and another on the top for reduction of the channel dimension (aka depth dimension).

To improve the accuracy of classification model for mitigating the negative effect of vanishing gradients caused by activation functions in a very deep network, skip-connection technique is deployed for associating M-blocks as described in Fig. 1. Specifically, the input of the current M-block x (as far as the output of the previous block) is combined with its own output via an element-wise addition. Formally, this process is formulated as follows

$$\mathsf{y} = \mathcal{R}(\mathsf{x}) + \mathsf{x}. \tag{5}$$

As a convincing explanation, the output of skip-connection scheme stands for a mapping function $\mathsf{y} = \mathcal{S}(\mathsf{x})$. Accordingly, instead of learning the true output $\mathcal{S}(\mathsf{x})$ in a traditional structure, the skip-connection mechanism allows M-blocks to learn the residual information $\mathcal{R}(\mathsf{x})$. The output y is continuously used for the next M-block with same skip-connection strategy. At the end of network, the feature maps of the last M-block are gathered with its input (aka the output of previous skip-connection) by a depth concatenation layer.

$$\mathsf{y}_{\mathsf{final}} = \mathrm{concat}\left(\mathcal{R}(\mathsf{y}_5), \mathsf{y}_5\right), \tag{6}$$

where y_5 refers to as the skip-connection output of the fifth M-block. Obviously, the diverse features extracted in each block and the informative identity maintained throughout the network via skip-connection scheme are jointly synthesized to enrich the classification model. As a result, MCNet is capable of being against the problems of vanishing gradients and overfitting during the network training phase. At the end of MCNet, a average-pooling layer, a fully connected layer, and a softmax layer are arranged sequentially after the concatenation layer. The number of hidden units C in the fully connected layer is set equal to the number of modulation classes of a given dataset. Regarding the network configuration reported in Table I, MCNet has 141,880 trainable parameters in total. The network capacity and computational complexity can be diminished by reducing the number of M-blocks as well as kernels in convolutional layers. However, the recognition accuracy will be downgraded because the network is not deep enough to learn strongly discriminative features at multi-scale representations of feature maps.

TABLE I
CONFIGURATION OF THE PROPOSED NETWORK ARCHITECTURE

Layer	**Output Volume**	**Description** (or remarks)
input	$2 \times 1024 \times 1$	
conv	$2 \times 512 \times 64$	$64 \times (3 \times 7)$
pool	$2 \times 256 \times 64$	max-pooling $(1, 3)$
pre-block	$2 \times 128 \times 64$	$32 \times (1 \times 3)$, $32 \times (3 \times 1)$
pool	$2 \times 64 \times 64$	max-pooling $(1, 3)$
M-block (p)	$2 \times 32 \times 128$	$48 \times (3 \times 1)$, $48 \times (1 \times 3)$ $2 \times [32 \times (1 \times 1)]$, max pooling $(1, 3)$
add	$2 \times 32 \times 128$	element-wise addition
M-block	$2 \times 32 \times 128$	$48 \times (3 \times 1)$, $48 \times (1 \times 3)$ $2 \times [32 \times (1 \times 1)]$
add	$2 \times 32 \times 128$	element-wise addition
M-block (p)	$2 \times 16 \times 128$	$48 \times (3 \times 1)$, $48 \times (1 \times 3)$ $2 \times [32 \times (1 \times 1)]$, max pooling $(1, 3)$
add	$2 \times 16 \times 128$	element-wise addition
M-block	$2 \times 16 \times 128$	$48 \times (3 \times 1)$, $48 \times (1 \times 3)$ $2 \times [32 \times (1 \times 1)]$
add	$2 \times 16 \times 128$	element-wise addition
M-block (p)	$2 \times 8 \times 128$	$48 \times (3 \times 1)$, $48 \times (1 \times 3)$ $2 \times [32 \times (1 \times 1)]$, max pooling $(1, 3)$
add	$2 \times 8 \times 128$	element-wise addition
M-block	$2 \times 8 \times 256$	$96 \times (3 \times 1)$, $96 \times (1 \times 3)$ $32 \times (1 \times 1)$, $64 \times (1 \times 1)$
concat	$2 \times 8 \times 384$	feature map concatenation in depth
pool	$1 \times 1 \times 384$	average-pooling $(2, 8)$
classification	$1 \times 1 \times \mathsf{C}$	fully-connected with C neurons softmax

III. NUMERICAL RESULTS AND DISCUSSION

A. Dataset Description

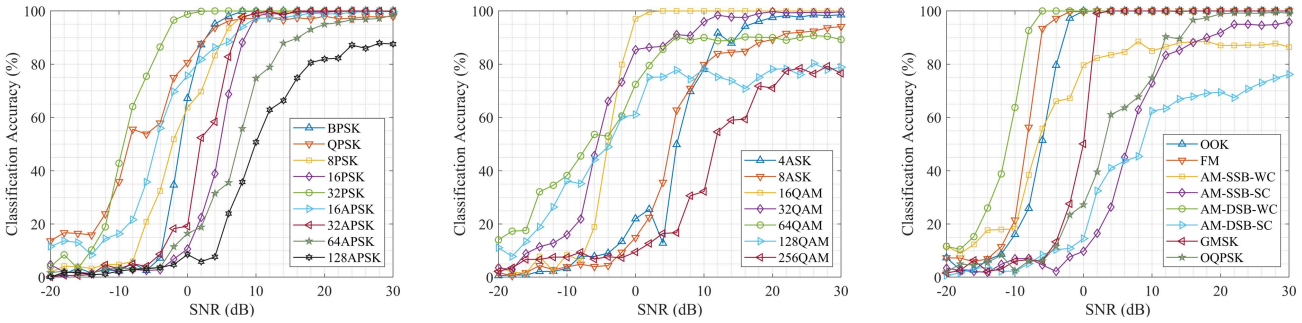
The proposed CNN-based AMC method is benchmarked on DeepSig: RadioML (version 2018.01A) [13], one of the most challenging datasets of modulation classification. The newest version includes both synthetic simulated channel effects (such as carrier frequency offset, symbol rate offset, delay spread, thermal noise) and over-the-air measurements of 24 digital and analog modulation formats spreading in a wide-ranging SNR (from -20 dB to $+30$ dB SNR). The dataset has over 2.5 million 1024-length frames of modulation signal, where 80% for training and the remainder for testing. The classification accuracy is reported on the test set.

B. Results and Analysis

As model training options, MCNet is end-to-end trained from scratch with randomly initial weights in 60 epochs using the stochastic gradient descent optimizer with the momentum of 0.9. The mini-batch size for each iteration is set to 128 and the learning rate is initialized at 0.01 (dropped 90% after 30 epochs). The performance is measured on a system equipped by a 3.70-GHz CPU, 16GB RAM, and a single NVIDIA GeForce GTX 1080Ti GPU.

In the first experiment, the classification accuracy of 24 modulations is fully reported in Fig. 3, wherein the results are grouped into three categorized for a better visualization, including PSK + APSK in Fig. 3a, ASK + QAM in Fig. 3b, and low-order + analog in Fig. 3c. Many PSK modulations reach the accuracy over 80% at $+0$ dB SNR (incredibly, 32PSK achieves approximately 98.89%). Meanwhile, MCNet recognizes 16QAM and 32QAM signals competently with the accuracy rates of 98.78% and 82.91%, respectively, at $+0$ dB SNR. The recognition rate keeps getting worse along the increment of APSK and QAM orders due to vulnerability of high-order modulation signal under various channel impairments. For instance, the accuracy significantly decreases

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(a) PSK+APSK (b) ASK+QAM (c) Low-order+analog

Fig. 3. Performance of 24 1024−length modulations categorized into three groups of modulations: (a) PSK + APSK, (b) ASK + QAM, and (c) low-order + analog.

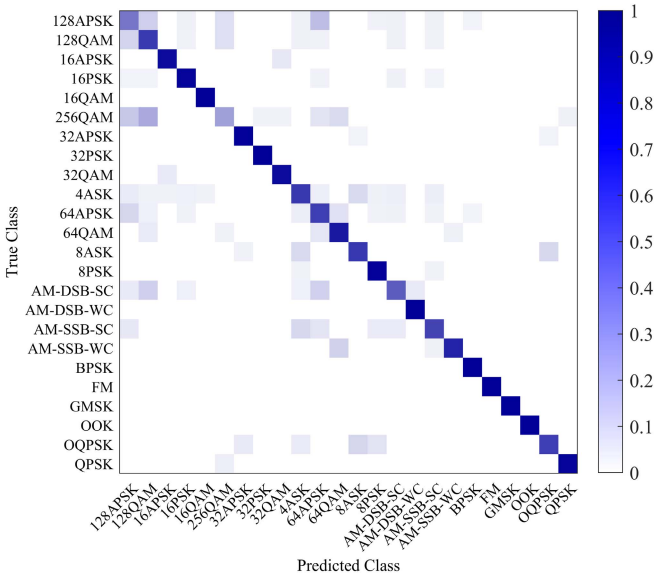


Fig. 4. Confusion matrix of 24-modulation classification at +10 dB SNR, where the accuracy is over 93%.

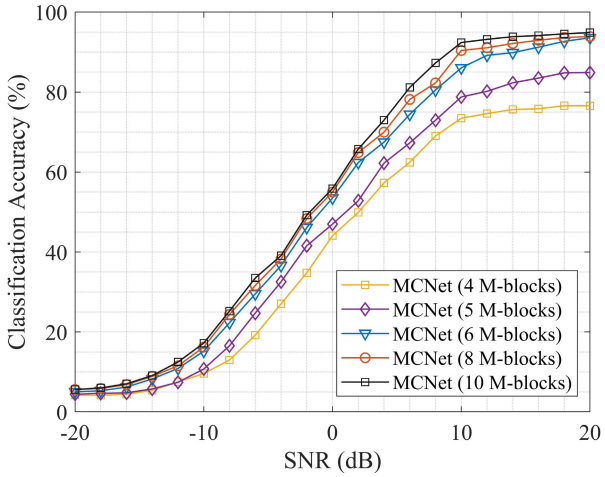


Fig. 5. Accuracy of 24-modulation classification with different number of M-blocks.

over 24% and 42% when upgrading the APSK order from 64 to 128 and the QAM order from 128 to 256, respectively. Some specific modulations with low information rate such as AM and FM are effortlessly recognized at low SNR regimes (surprisingly, AM-DSB-WC yields 100% rate at −6 dB SNR). In details, the confusion matrix of 24-modulation classification at +10 dB SNR is extra given in Fig. 4 for an insightful analysis, in which some modulations are confused by the others. From the confusion matrix, several low-order modulations such as FM and BPSK achieve remarkable recognition result with less confusion and so far no error. On the contrary, 128APSK, which gets the poor accuracy of 51.40% at +10 dB SNR, is mostly confused with 64APSK and 128QAM. Being reported as the worst modulation, 256QAM strongly suffers the confusion with 128APSK and 128QAM. With respect to high-order modulations, it is possible to achieve a high transmission rate, but the modulation recognition of a radio signal becomes more challenging with the error rate raised by the fact that closer constellation points are vulnerable with noise. In order to investigate the efficiency of convolutional blocks, the performance of MCNet is measured by varying the number of M-blocks. Based on the recognition results plotted in Fig. 5, more M-blocks the MCNet architecture has, more accurately the modulations are classified. Compared with the default architecture that adopts six M-blocks, two diminished versions with four and five M-blocks perform significantly worse due to the lack of deep features at multi-scale representative maps, for instance, at the output volume of 2×8 and 2×16. On the other hand, involving more M-blocks only diversifies the features at the volume of 2×8, hence the performance is improved slightly at high SNR regimes.

In the second experiment, MCNet is compared with a baseline method [13], a VGG-based neural network [13], a residual neural network (ResNet) [13] on DeepSig, and a CNN-AMC network [8], where the comparison result is plotted in Fig. 6. The baseline with an advanced decision tree classifier presents the worst accuracy. With the high-relevant information acquired by residual architecture, ResNet outperforms the baseline, CNN-AMC, and VGG by over 9.00% at high SNR regimes. Although sharing the same backbone architecture, VGG is marginally better than CNN-AMC thanks to the deployment of more 3×3 kernels at every convolutional layer for proficiently dealing with a larger number of

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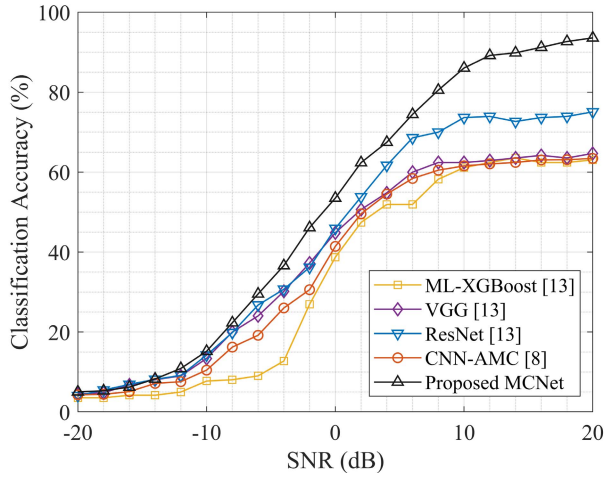


Fig. 6. Method comparison in terms of 24-modulation classification accuracy.

TABLE II
COMPARISON OF COMPUTATIONAL COMPLEXITY

Network	Capacity (No. parameters)	Inference time (ms)
VGG [13]	257K	0.131
ResNet [13]	236K	0.146
CNN-AMC [8]	575K	0.127
MCNet (4 M-blocks)	103K	0.095
MCNet (5 M-blocks)	120K	0.115
MCNet (6 M-blocks)	142K	0.125
MCNet (8 M-blocks)	180K	0.129
MCNet (10 M-blocks)	220K	0.131

modulation classes. Meanwhile, MCNet defeats the baseline and two deep CNN models with a significant improvement of accuracy, for instance, better than VGG and ResNet by 23.67% and 12.4%, respectively, at +10 dB SNR. For the computational complexity analysis, the network capacity and the average inference time (aka the prediction time) are summarized in Table II. MCNet is cheaper than ResNet and VGG by approximately 40% and 45% of capacity (aka the number of trainable parameters), respectively. Being designed with three fully connected layers, CNN-AMC is significantly bigger than ResNet and VGG, however, it performs more quickly because of fewer convolution operations. Despite being more compact than CNN-AMC, the inference time of MCNet is nearly equivalent to that of CNN-AMC due to several addition operations and depth-wise concatenation operations performed throughout the network. Additionally, it is worth noting that MCNet involving more M-blocks has to require more trainable parameters and be subject to longer inference time.

IV. CONCLUSION

In this letter, a cost-efficient and high-performed CNN, namely MCNet, is proposed to effectively classify the modulations of radio signal under various impairments of wireless channel. With various asymmetric kernels arranged in parallel in each convolutional block, the network is capable of fully capturing the spatiotemporal signal correlations. By leveraging skip connections in block-wise manner throughout the network, overfitting and gradient vanishing are mostly prevented during the training process. In the experiments, MCNet reaches the overall classification accuracy of over 93% at 20 dB SNR on the DeepSig dataset of 24 challenging modulations and significantly outperforms both the traditional ML-based and innovative DL-based models.

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